

Major Front-End Migration for GRNsight, a Web Application for Visualizing Gene Regulatory and Protein-Protein Interaction Network Models

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Undergraduate Research Conference
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Loyola Marymount University
Frank R. Seaver
College of Science
and Engineering

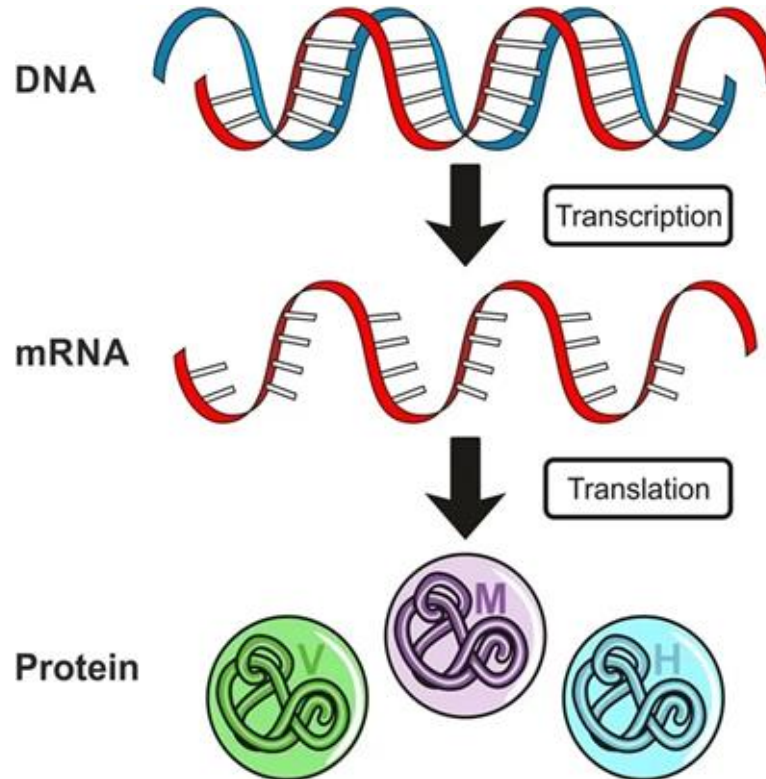
Outline

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein-Protein Interactions (PPI).
- GRNsight is an integral part of a data analysis workflow for biologists who model GRNs.
- GRNsight's interface was originally built using older web tools, leading to repeated code and difficult user interface updates.
- I have migrated GRNsight's user interface to React Vite and Grommet UI which are modern, scalable, and industry-standard frameworks.
- GRNsight React migration has resulted in user interface improvements and bug fixes that enhance the user experience for biologists.

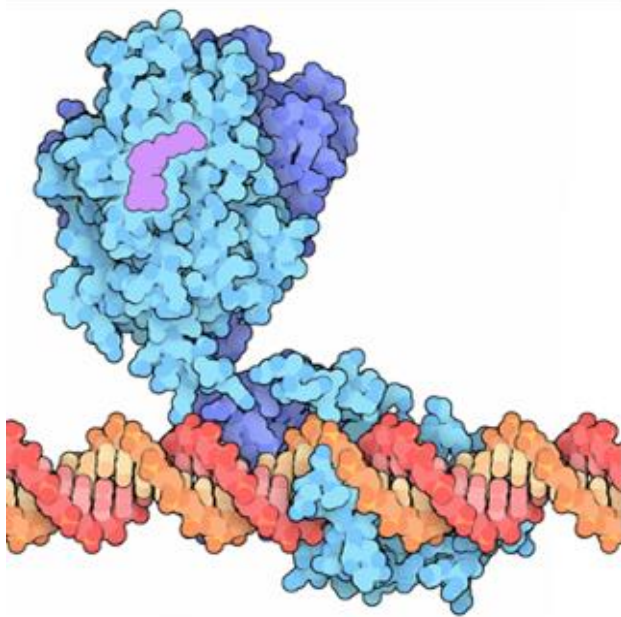
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Gene expression is the flow of information in a cell from DNA to RNA to protein



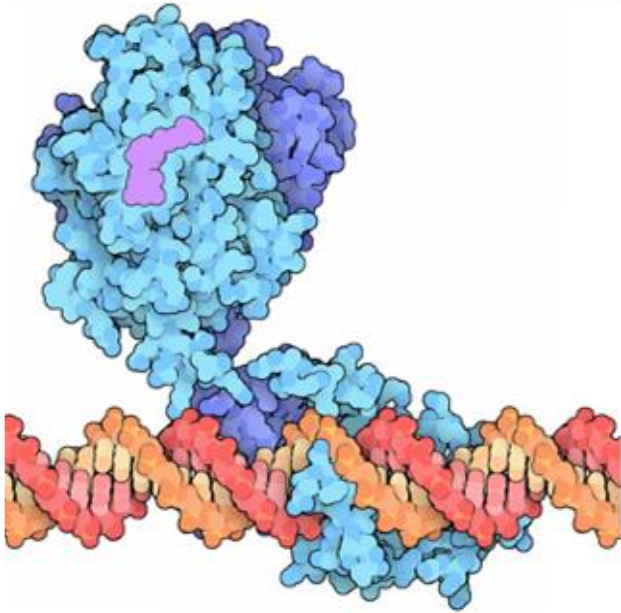
Regulatory Transcription Factors bind to DNA and turn on (activate) or turn off (repress) gene expression



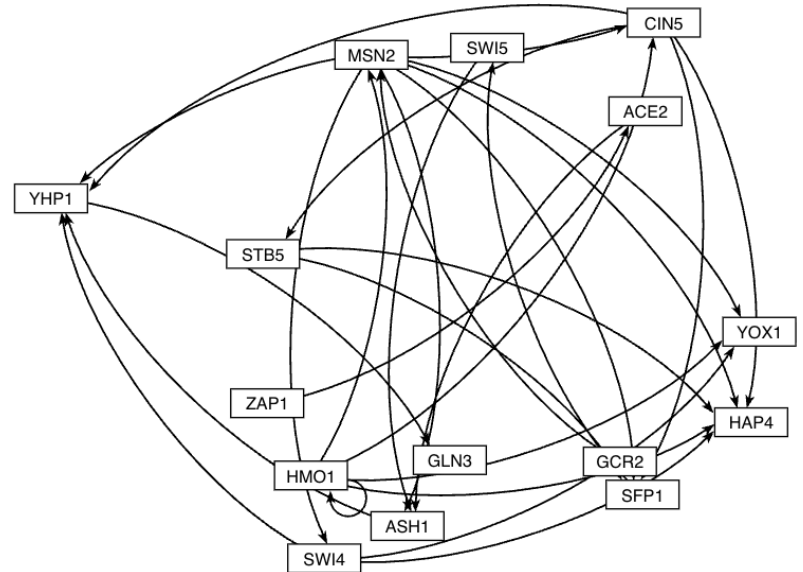
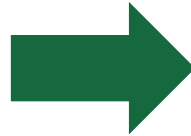
- In yeast ~250 regulatory transcription factors regulate ~6000 genes.
- In humans, ~500 regulatory transcription factors regulate ~20,000 genes

"PDB101: Molecule of the Month: Vitamin D receptor,"
RCSB: PDB-101. <https://pdb101.rcsb.org/motm/155>

A Gene Regulatory Network (GRN) consists of genes, transcription factors and the regulatory connections between them which govern the level of expression of mRNA and protein from genes



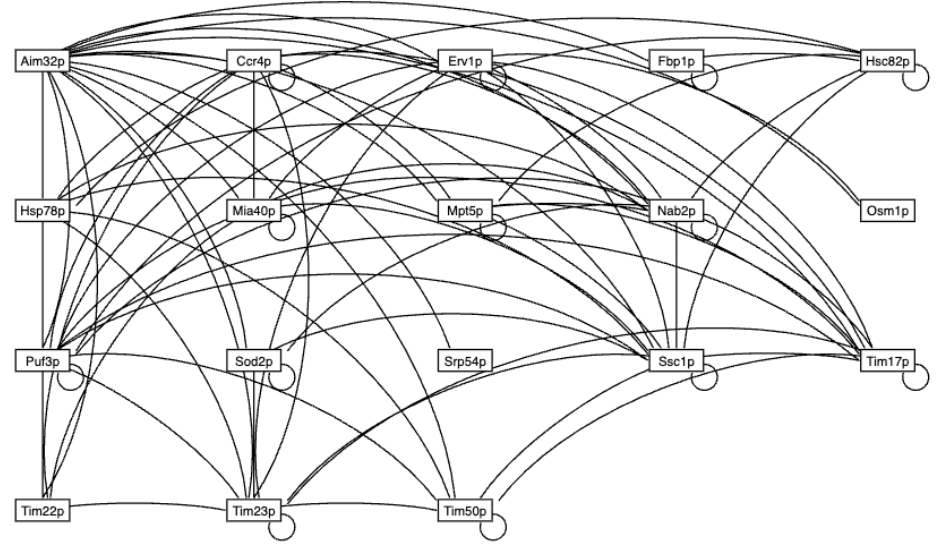
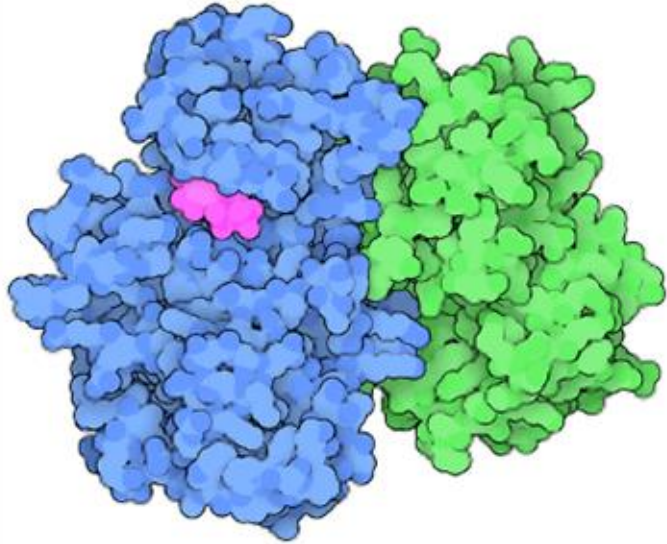
"PDB101: Molecule of the Month: Vitamin D receptor,"
RCSB: PDB-101. <https://pdb101.rcsb.org/motm/155>



Yeast GRN that controls the cold shock response

- Node = transcription factor / gene
- Arrow edge = regulatory relationship

Protein-Protein Interactions refer to physical interactions of proteins binding to each other - the basis of nearly all cell functions



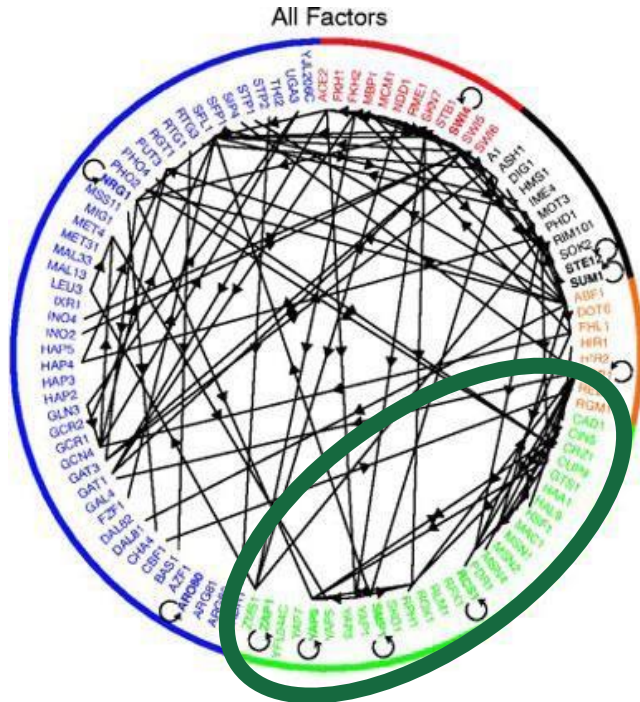
"PDB101: Molecule of the Month: Cyclin and cyclin-dependent kinase,"
RCSB: PDB-101. <https://pdb101.rcsb.org/motm/236>

PPI network for yeast mitochondrial protein Aim32p

- Node = proteins
- Edge = binding relationship

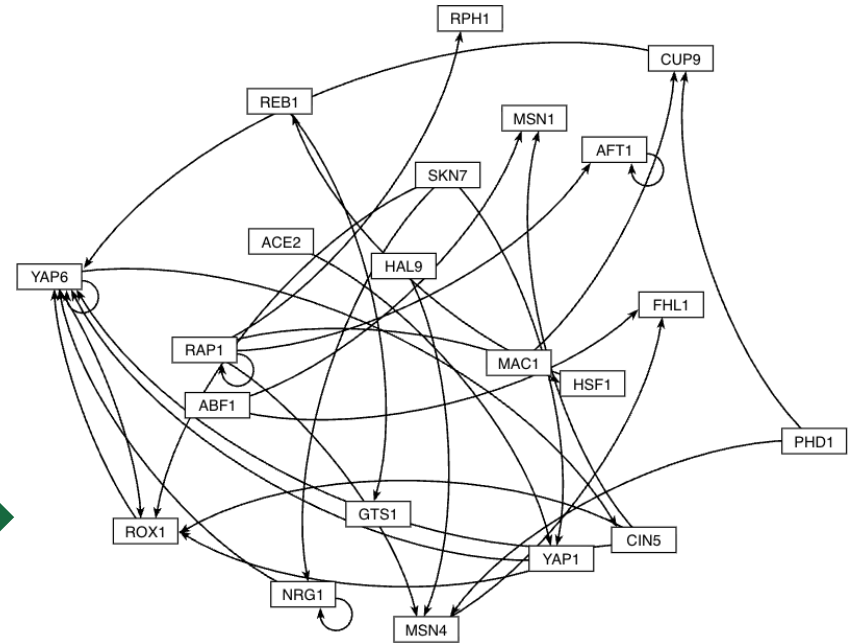
Large gene regulatory networks are difficult to visualize, and GRNsight makes it easier to visualize networks of 35 nodes or fewer

GRN of 106 transcription factors in yeast



Lee *et al.* (2002)

GRN subnetwork of 21 transcription factors that regulate the environmental stress response in yeast

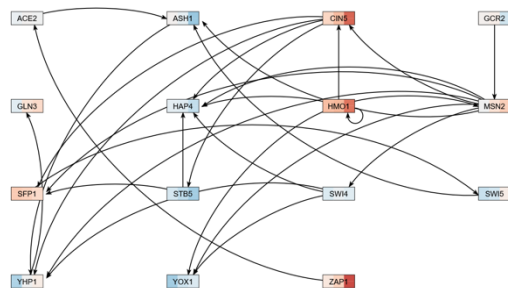


GRNsight

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GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Select network and
expression data from
GRNsight database

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type:

Network Source:

Select gene:

Add new genes go below! Click on a gene to remove it.

ACE2 (YAL001212)	ASH1 (YAL004006)	CINS (YAL004002)
GCR2 (YAL004002)	GLND (YAL004002)	HAP4 (YAL004006)
HMO1 (YAL004006)	MGN2 (YAL004006)	SFP1 (YAL004006)
STB5 (YAL004006)	SWH (YAL004006)	SWI5 (YAL004006)
YHP1 (YAL004006)	YOX1 (YAL004006)	

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded
expression data, or use data
from our Expression Database

Top Dataset

☒ Average Replicate Values

Bottom Dataset

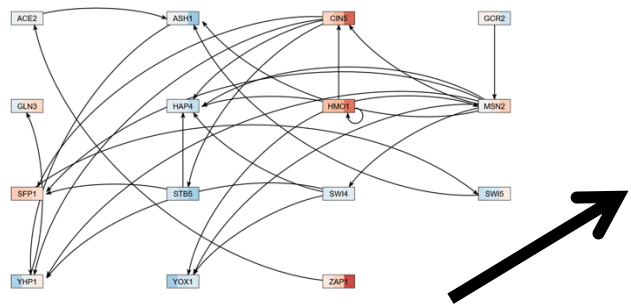
☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

Set

-3.50 0 3.50

GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Export to Excel



Cell	Value
ACE2	1
ASH1	1
CINS	1
GCR2	1
GND	1
HAP4	1
HMO1	1
MSH2	1
SFP1	1
STB5	1
SWH	1
SWH5	1
YHP1	1
YOK1	1
ZAP1	1



Select network and expression data from GRNsight database

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type:

Network Source:

Select gene:

Add new genes go below! Click on a gene to remove it.

ACE2
(YAL0012C)

ASH1
(YAL0012C)

CINS
(YAL0012C)

GCR2
(YAL0012C)

GND
(YAL0012C)

HAP4
(YAL0012C)

HMO1
(YAL0012C)

MSH2
(YAL0012C)

SFP1
(YAL0012C)

STB5
(YAL0012C)

SWH
(YAL0012C)

SWH5
(YAL0012C)

YHP1
(YAL0012C)

YOK1
(YAL0012C)

ZAP1
(YAL0012C)

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

☒ Average Replicate Values

Bottom Dataset

☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

Set

-3.50 0 3.50

Export to Excel

Select the Expression Data Source:

☐ None

☐ Agilent_2012

☐ Baranyi_2012

☐ Dahquist_2018

☐ Klingmann_2002

☐ Thomas_2007

Select Workbook Sheets to Export:

☒ Select All

Network Sheets:

☒ network

☐ network_optimized_weights

Expression Sheets:

☒ dms0_log2_expression

☒ dms1_log2_expression

☒ dms2_log2_expression

☒ wt_log2_expression

Additional Sheets:

☒ degradation_rates

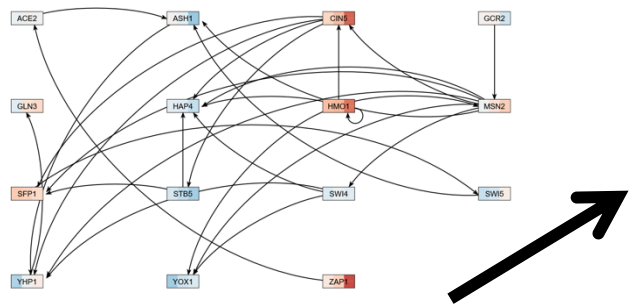
☒ optimization_parameters

☒ production_rates

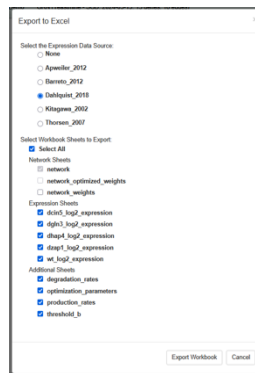
☒ threshold_b

Export Workbook Cancel

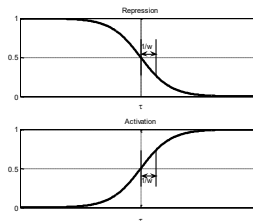
GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Export to Excel



Dynamical systems modeling using GRNmap



Select network and expression data from GRNsight database

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type: Gene Regulatory

Network Source: Allanoreline - SGD: 2006-01-22

Select gene: HAP4

Add new genes go below! Click on a gene to remove it.

ACE2 (YJL017C)	ASH1 (YJL009C)	CRN5 (YJL009C)
GCR2 (YJL009C)	GND2 (YJL009C)	HAP4 (YJL009C)
HMO1 (YJL009C)	MND2 (YJL009C)	SFP1 (YJL009C)
STB5 (YJL009C)	SWH4 (YJL009C)	SWH5 (YJL009C)
YHP1 (YJL009C)	YOK1 (YJL009C)	ZAP1 (YJL009C)

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

Dahlquist_2018_wt

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

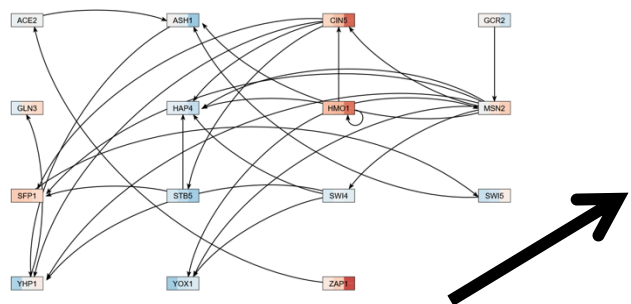
Log Fold Change Max Value
(0.01-100):

3

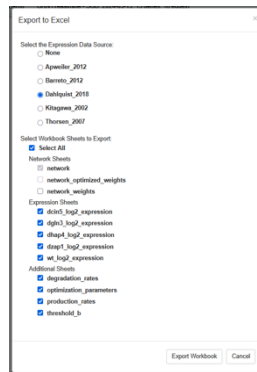
Set

-3.50 0 3.50

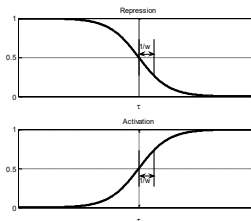
GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



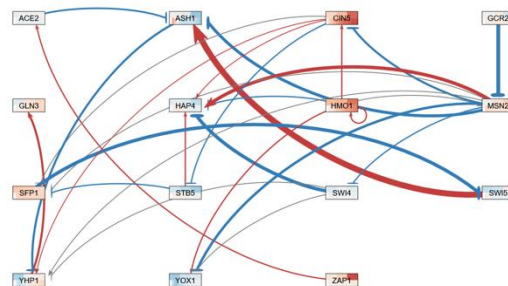
Export to Excel



Dynamical systems modeling using GRNmap



Visualization of modeling results using GRNsight



Select network and expression data from GRNsight database



Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below.

Network Type: Gene Regulatory

Network Source: AllanoreMine - SGD 2006-01-22

Select gene: DAPE

Add new genes go below! Click on a gene to remove it.

ACE2 (YAL0072C)	ASH1 (YAL0080C)	CINS (YAL0082C)
GCR2 (YAL0084C)	GLN3 (YAL0086C)	HAP4 (YAL0088C)
HMO1 (YAL0090C)	MSN2 (YAL0092C)	SFP1 (YAL0094C)
STB5 (YAL0096C)	SWH (YAL0098C)	SWI5 (YAL0100C)
YHP1 (YAL0102C)	YOX1 (YAL0104C)	ZAP1 (YAL0106C)

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

Dahlquist_2018_wt

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

3

Set

-3.50 0 3.50

Users can import networks via the file upload functionality in a supported format (SIF, GraphML, or Excel)



GRNsight

Web app for visualizing models of small-scale gene regulatory and protein-protein interaction networks.

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Welcome to GRNsight. Please select a network to view.

[Network](#)[Layout](#)[Node](#)[Edge](#)[View](#)[Export](#)[Help](#)[Demo](#)

Network

Network Source
Demo

Select a Demo ▾

 Open File

 Load from Database

 Reload

Network Mode:
No Network Selected

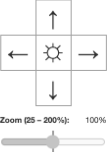
Species:
Saccharomyces cerevisiae

Layout

Node

Edge

View



Zoom (25 - 200%): 100%

Users can load networks from our database, which contains regulation data from the *Saccharomyces* Genome Database (SGD)



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Network Layout Node Edge View Export Help Demo

Network

Network Source

Demo

Select a Demo ▼

Open File

Load from Database

Reload

Network Mode:

No Network Selected

Species:

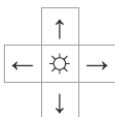
Saccharomyces cerevisiae

Layout

Node

Edge

View



Zoom (25 - 200%): 100%

Users can load different network types, either GRN or PPI



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Welcome to GRNsight. Please select a network to view.

Users can choose different network sources since GRNsight refreshes with new data each year



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GRNsight rapidly generates GRN graphs using our customizations to the open source D3 library



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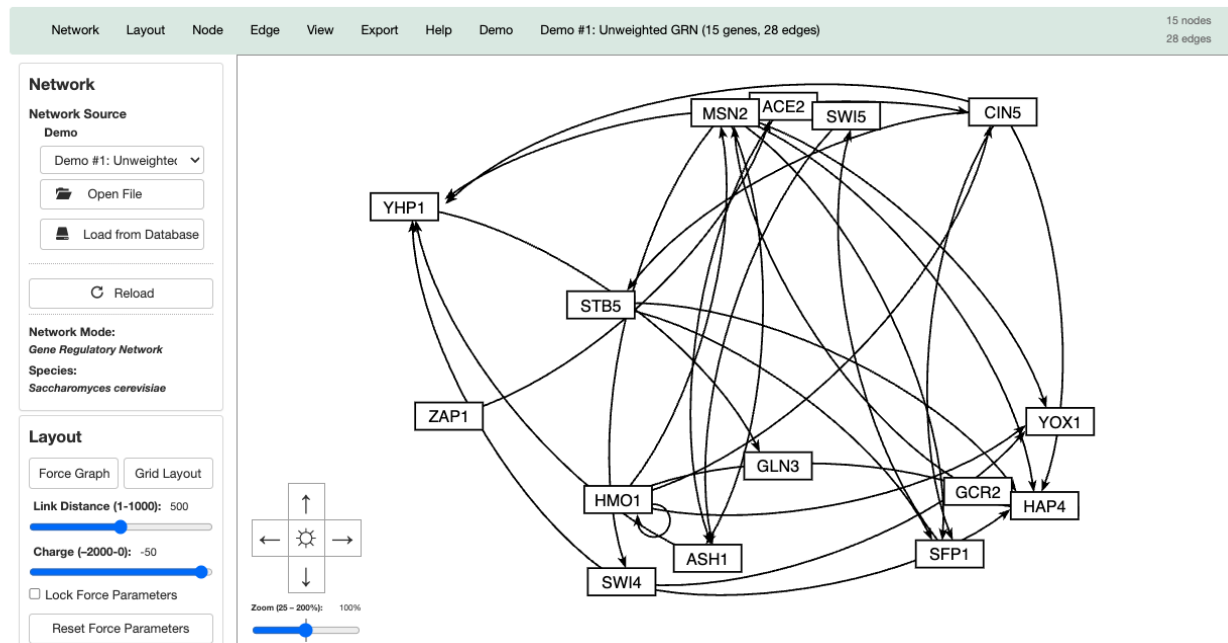
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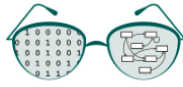
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Besides the Force Graph Layout, Users Can Rearrange the Nodes by Hand



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Welcome to GRNsight. Please select a network to view.

Network Layout Node Edge View Export Help Demo Demo #1: Unweighted GRN (15 genes, 28 edges)

15 nodes
28 edges

Network

Network Source

Demo

Demo #1: Unweighted

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

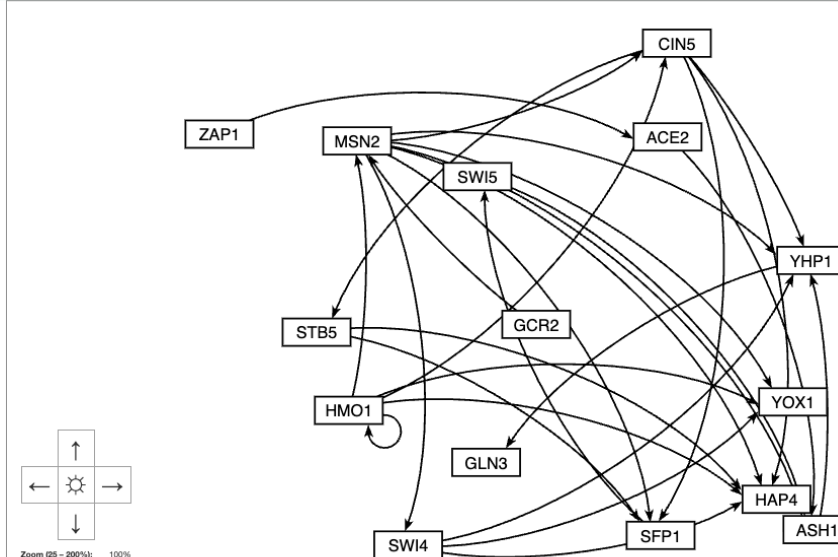
Force Graph Grid Layout

Link Distance (1-1000): 500

Charge (-2000-0): -50

☐ Lock Force Parameters

Reset Force Parameters



Or Choose the Convenient Block Layout Option



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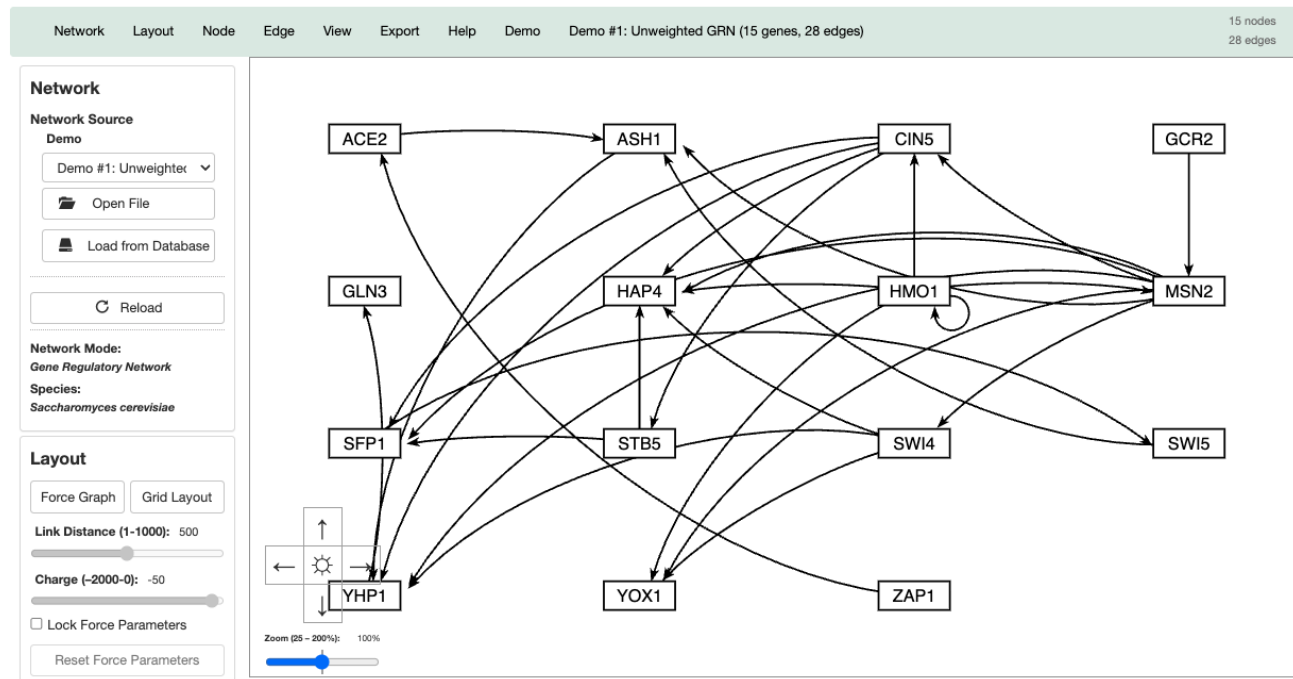
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When Model Results are Loaded, Edge Weight Values Are Indicated by Color, Thickness, and End Marker



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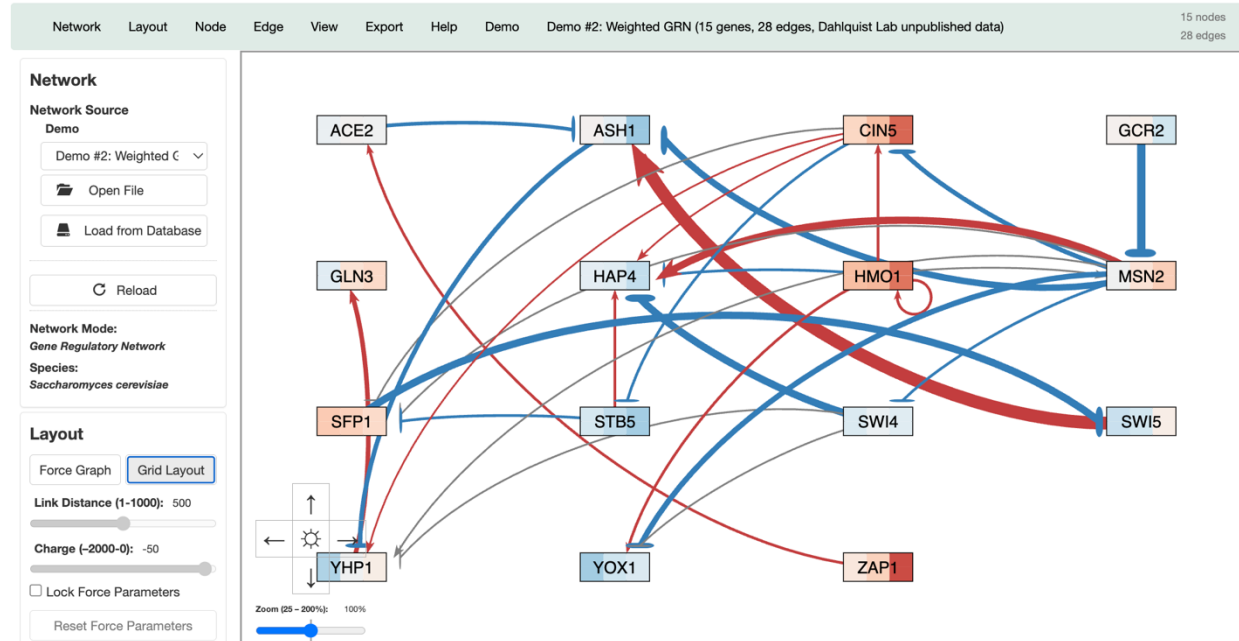
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Numerical Weight Values Can Be Displayed on the Graph



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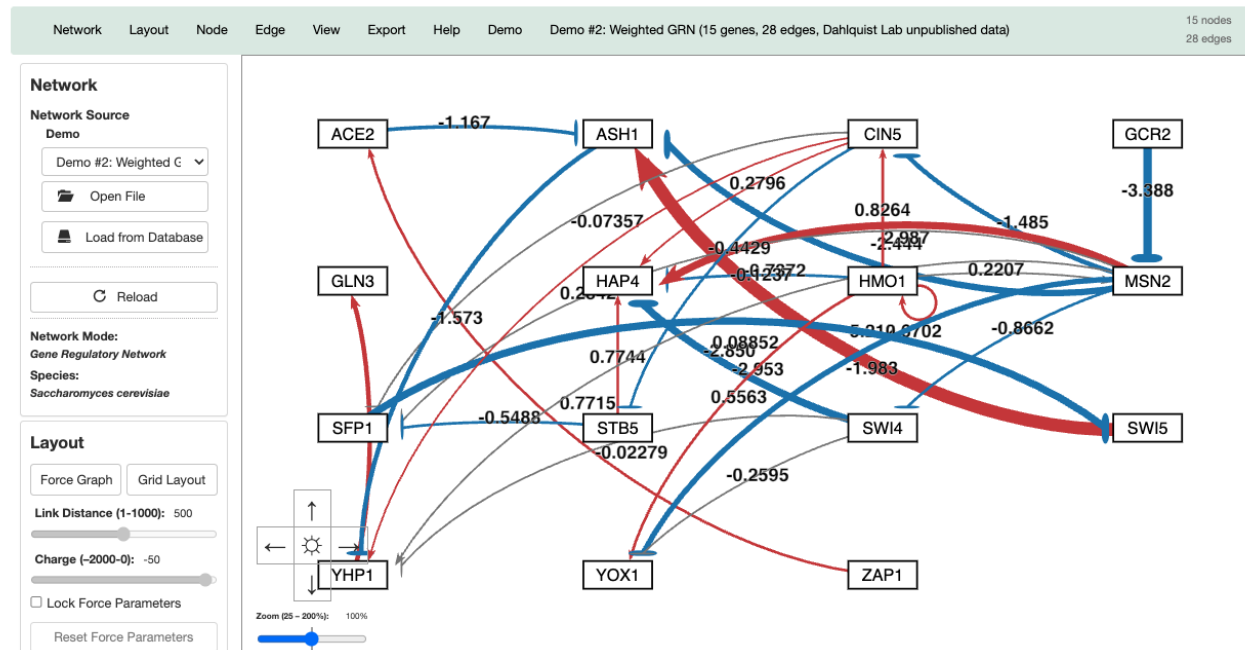
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Edge Thicknesses are Normalized to the Largest Magnitude Weight Value



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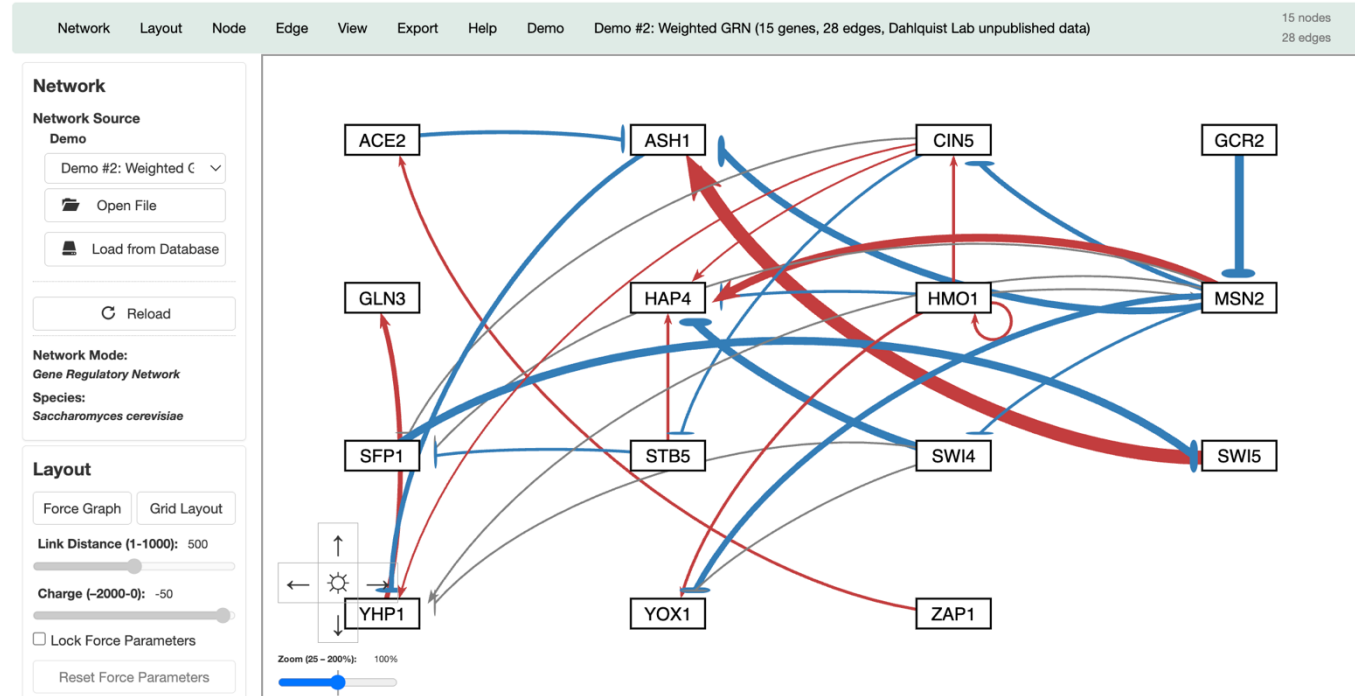
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Normalization Factor = 5.21



And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



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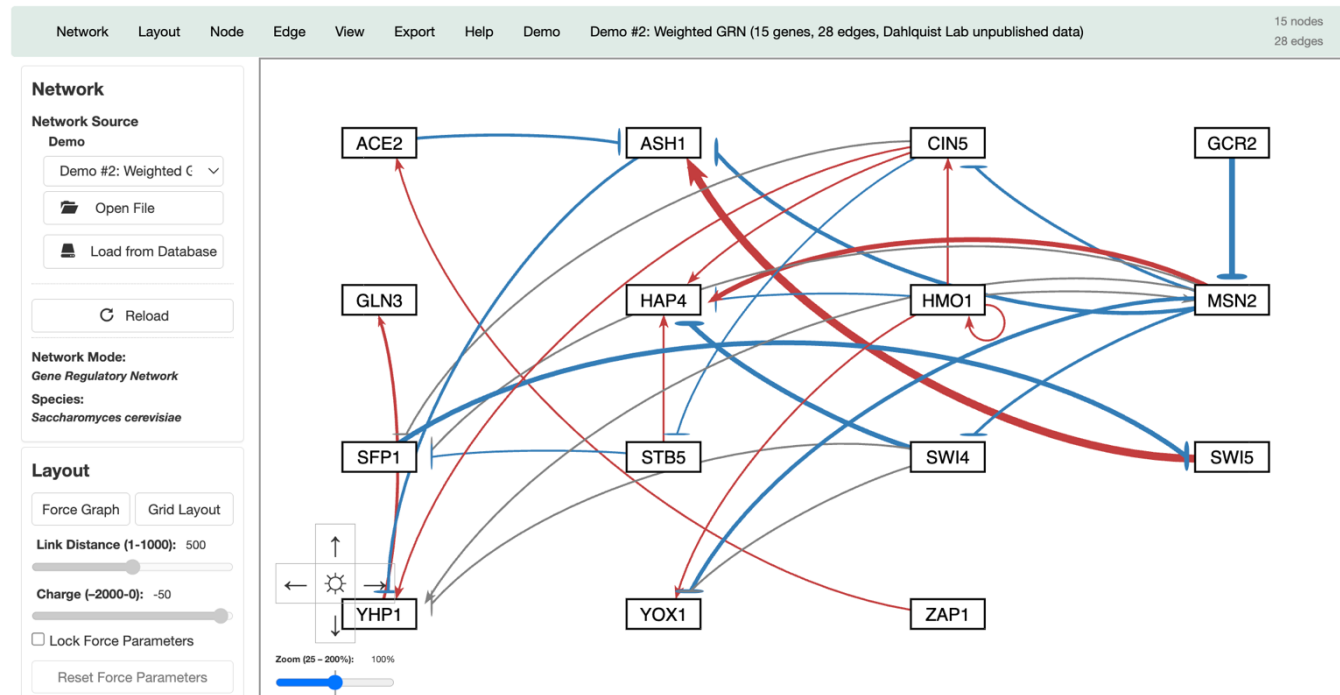
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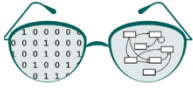
GRNsight v7.6.0

Normalization Factor = 10

Welcome to GRNsight. Please select a network to view.



And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



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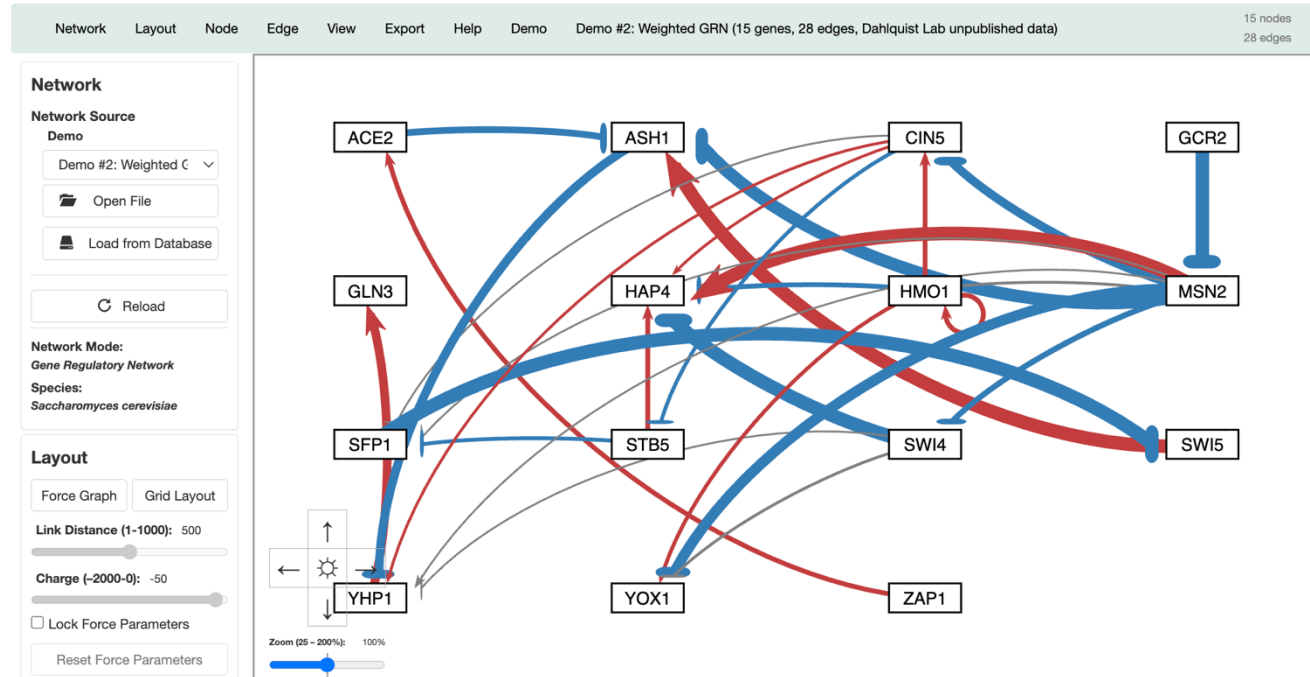
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Normalization Factor = 2.5

Welcome to GRNsight. Please select a network to view.



Edges with Normalized Weight Values within 5% of 0 Are Displayed as Gray to Minimize Their Visual Strength



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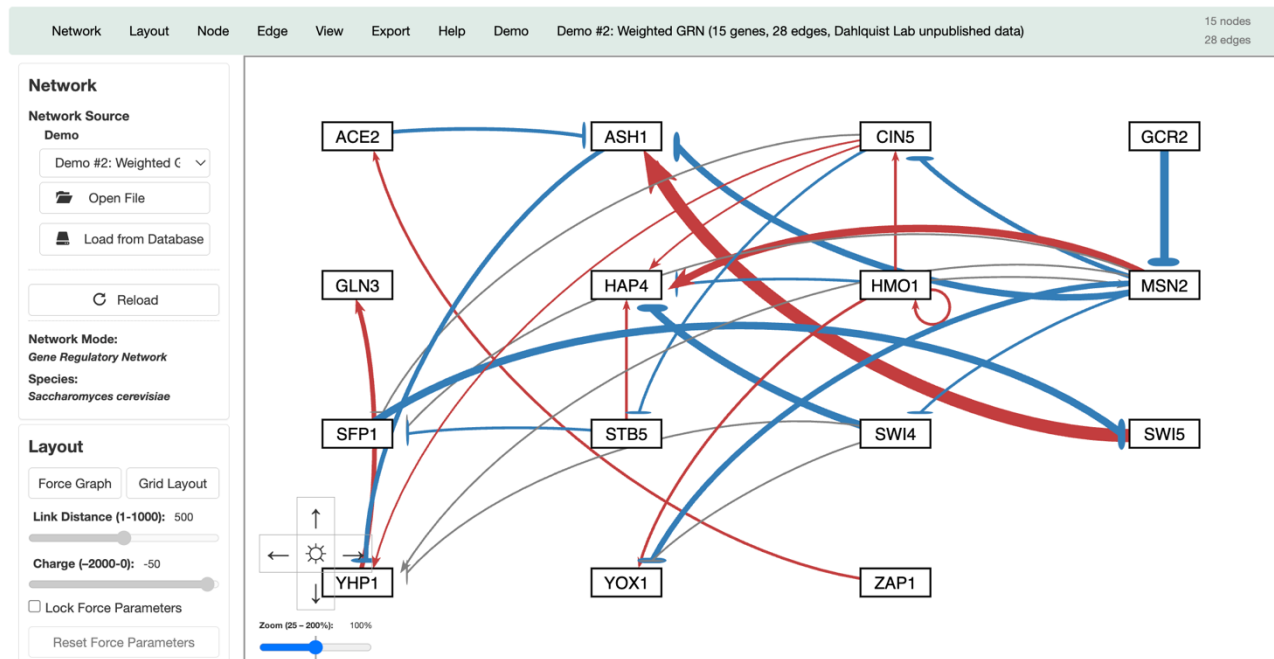
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GRNsight v7.6.0

Welcome to GRNsight. Please select a network to view.

Normalization Factor = 5.21
Gray Threshold = 5%



Users Can Change the Gray Threshold with a Slider



GRNsight

Web app for visualizing models of small-scale gene regulatory and protein-protein interaction networks.

A Joint Project of the
LMU Bioinformatics and
Biomathematics Groups

LMU|LA
Loyola Marymount
University

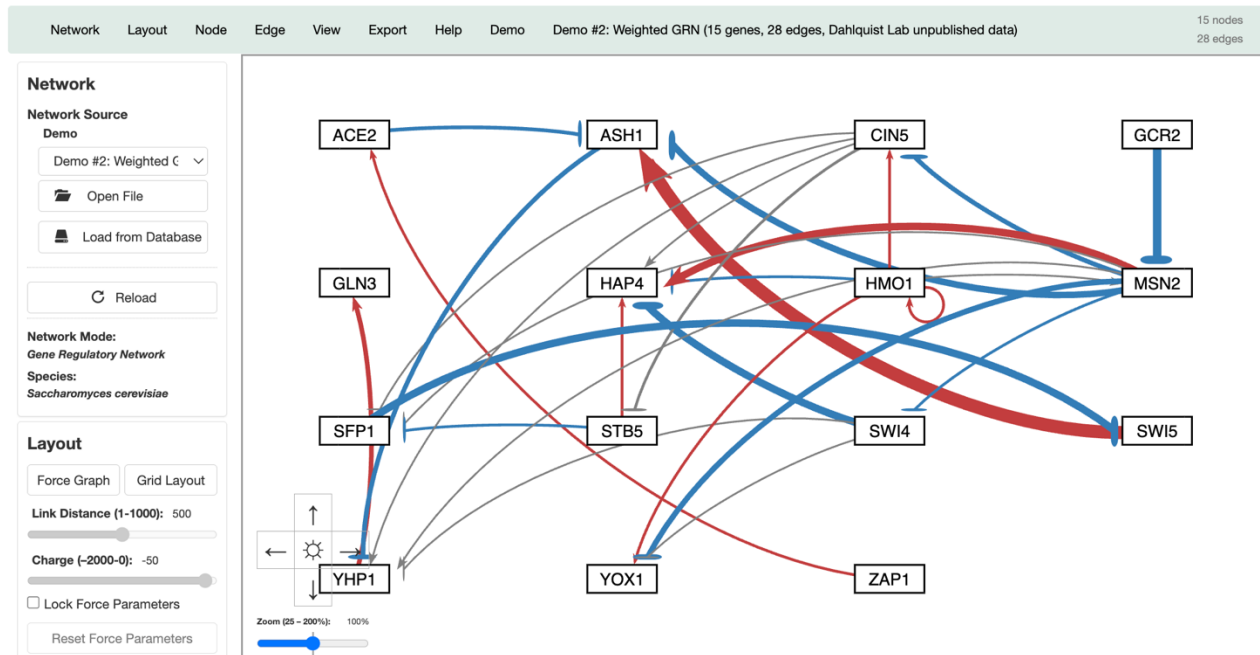
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3/18/2026

GRNsight v7.6.0

Welcome to GRNsight. Please select a network to view.

Normalization Factor = 5.21
Gray Threshold = 10%



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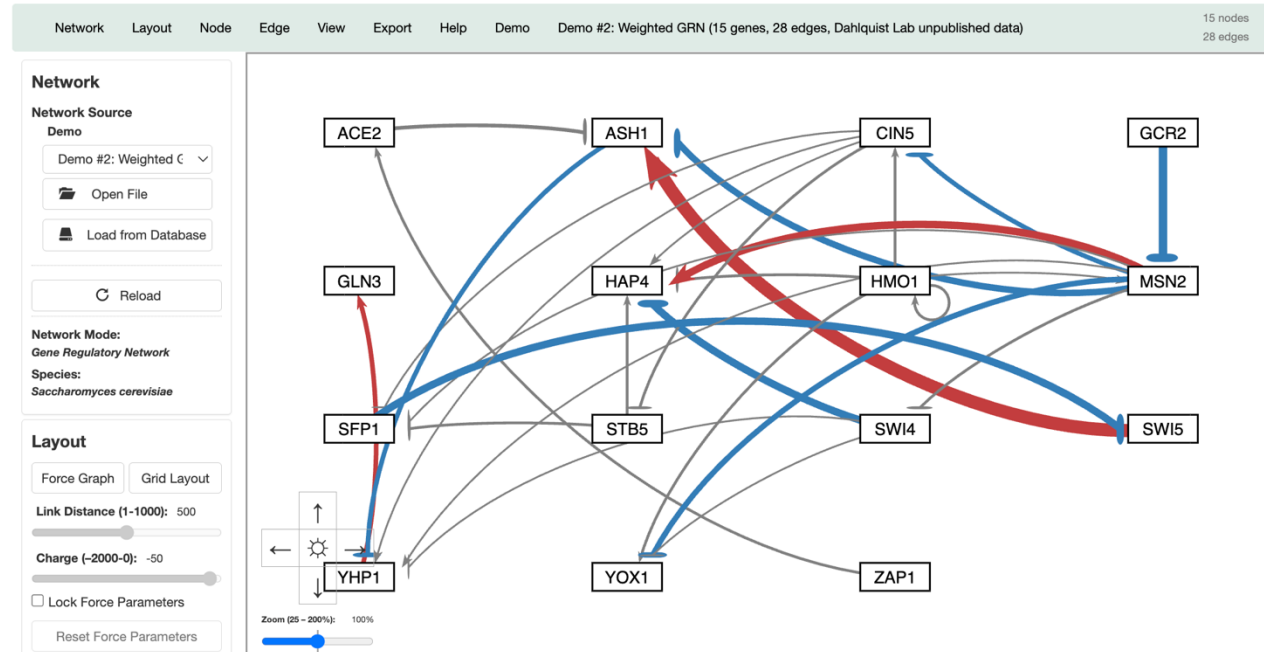
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GRNsight v7.6.0

Welcome to GRNsight. Please select a network to view.

Normalization Factor = 5.21
Gray Threshold = 25%



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Welcome to GRNsight. Please select a network to view.

Normalization Factor = 5.21
Gray Threshold = 50%

Network Layout Node Edge View Export Help Demo Demo #2: Weighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

15 nodes
28 edges

Network Source

Demo

Demo #2: Weighted C

 Open File Load from Database

 Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force Graph

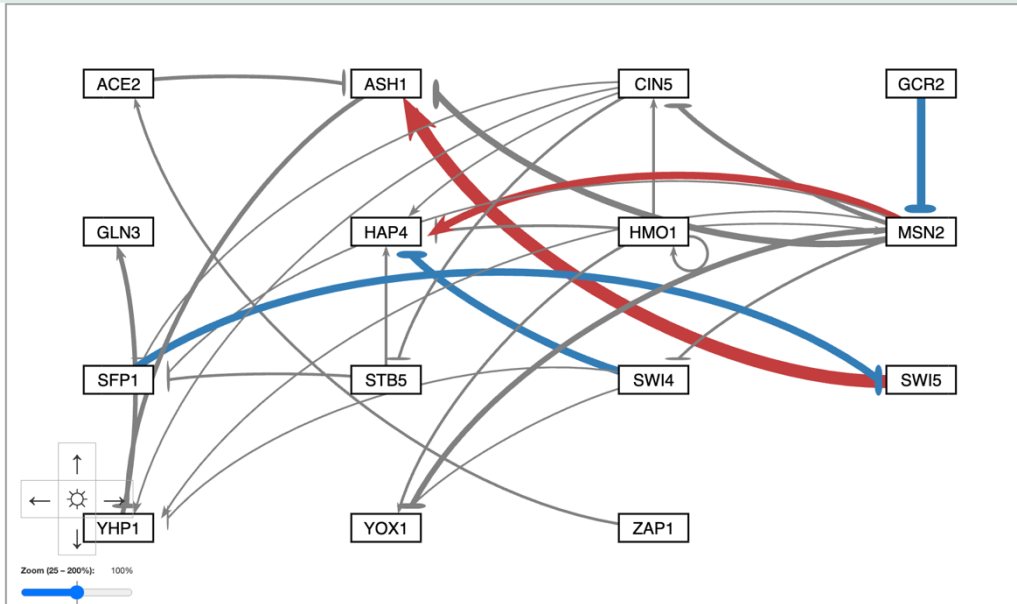
Grid Layout

Link Distance (1-1000): 500

Charge (-2000-0): -50

☐ Lock Force Parameters

Reset Force Parameters



Gray Edges Can Also Be Shown as Dashed Lines to Facilitate Visualization by Color Blind Users



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GRNsight v7.6.0

Welcome to GRNsight. Please select a network to view.

Network Layout Node Edge View Export Help Demo Demo #2: Weighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

15 nodes
28 edges

Network

Network Source

Demo

Demo #2: Weighted C

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force Graph Grid Layout

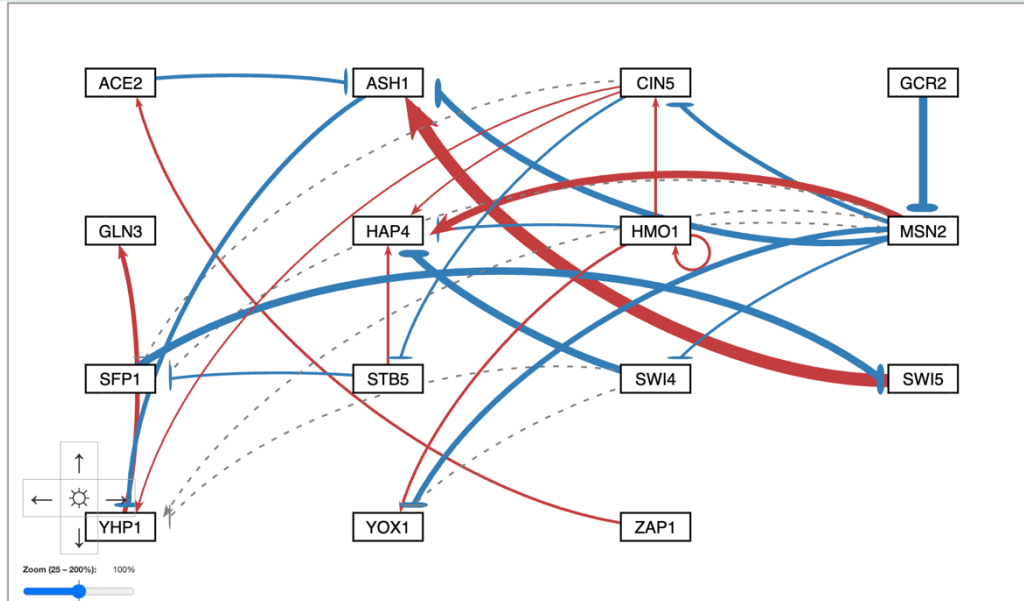
Link Distance (1-1000): 500

Charge (-2000-0): -50

Lock Force Parameters

Reset Force Parameters

Normalization Factor = 5.21
Gray Threshold = 5%



Nodes Can Display the Measured and/or Simulated Expression Data Uploaded by the User as Individual Heat Maps



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GRNsight v7.6.0

Welcome to GRNsight. Please select a network to view.

Network Layout Node Edge View Export Help Demo Demo #2: Weighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)

15 nodes
28 edges

Network

Network Source

Demo

Demo #2: Weighted G

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force Graph Grid Layout

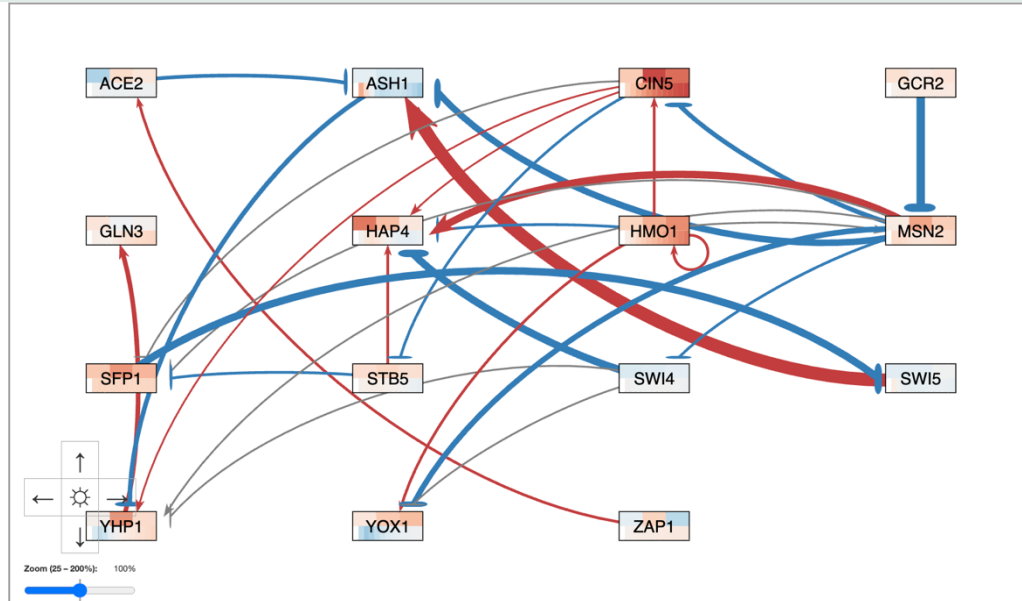
Link Distance (1-1000): 500

Charge (-2000-0): -50

Lock Force Parameters

Reset Force Parameters

Top = wild type yeast microarray data (Dahlquist lab)
Bottom = wild type yeast model simulated data



Additional datasets from a backend PostgreSQL expression database can be used to color nodes



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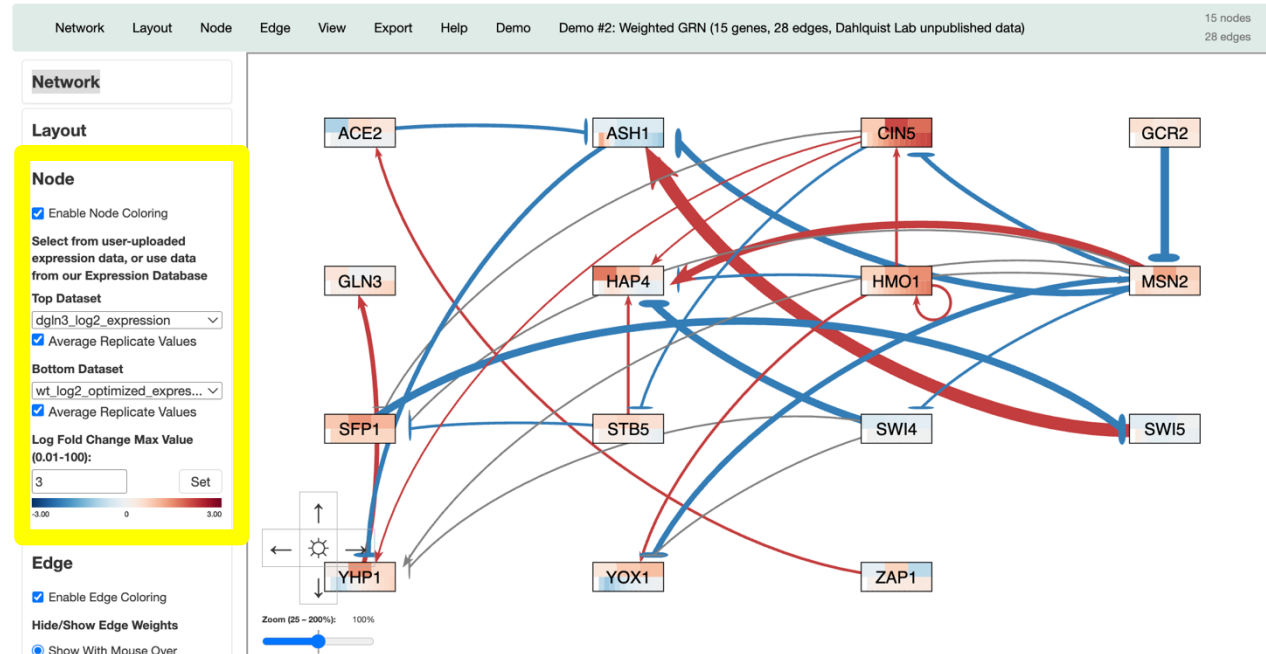
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GRNsight v7.6.0

Welcome to GRNsight. Please select a network to view.



Network and expression data can be exported to a GRNmap-compatible Excel Workbook for modeling



GRNsight

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GRNsight v7.2.0

Welcome to GRNsight. Please select a file to upload.

NetworkLayoutNodeEdgeViewExportHelpDemo

Demo #2: Weighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data)15 nodes28 edges

Network
Demo Source
Demo #2: Weighted GR ▾
Open File
Load from Database
Reload
Network Mode
Gene Regulatory Network ▾
Species:
Saccharomyces cerevisiae

Layout
Force GraphGrid Layout
Link Distance (1-1000): 500
Charge (-2000.0): -50
Lock Force Parameters

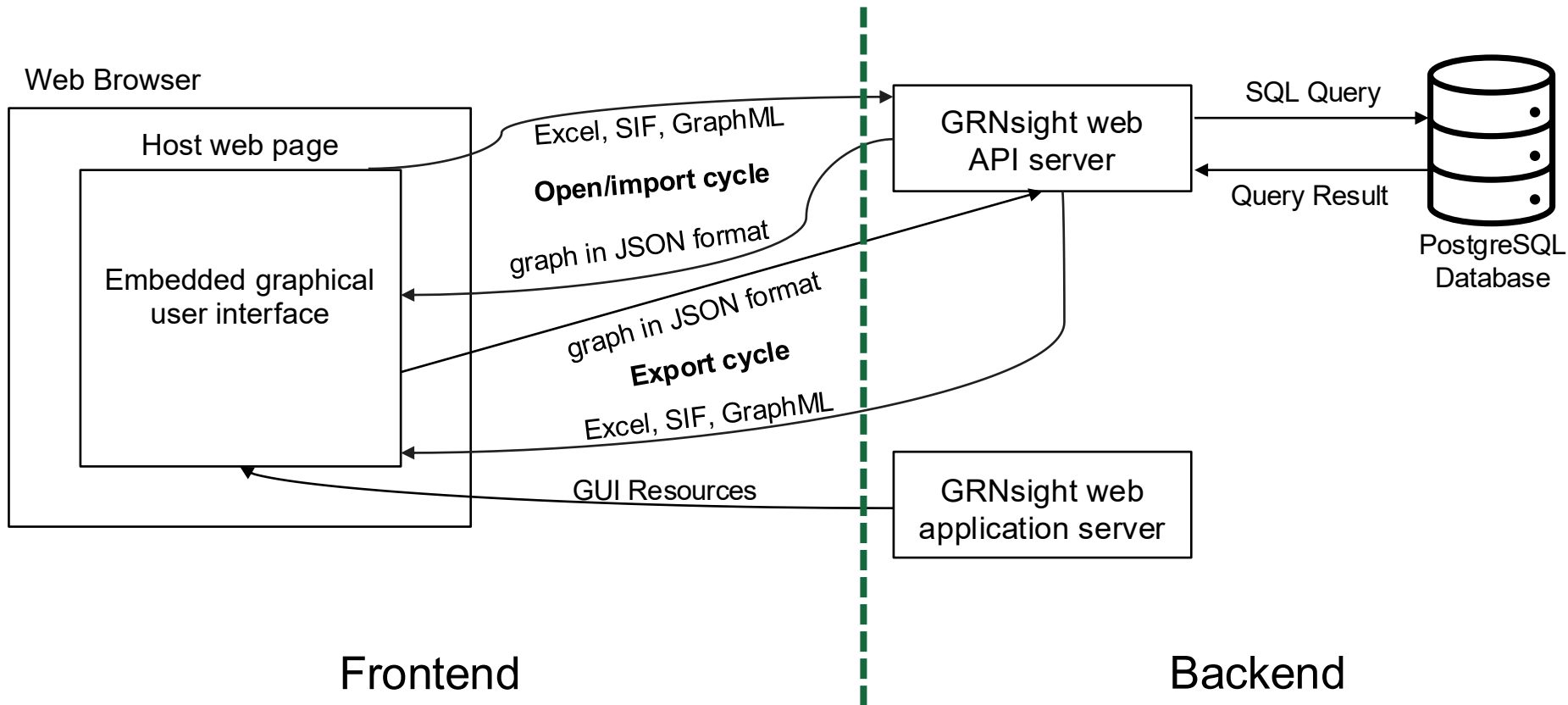
Export Data
Export Image
Print

To Unweighted SIF
To Weighted SIF
To Unweighted GraphML
To Weighted GraphML
To Excel

Outline

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Network (GRN) and Protein-Protein Interactions (PPI).
- GRNsight is an integral part of a data analysis workflow for biologists who model GRNs.
- GRNsight's interface was originally built using older web tools, leading to repeated code and difficult user interface updates.
- I have migrated GRNsight's user interface to React Vite and Grommet UI which are modern, scalable, and industry-standard frameworks.
- GRNsight React migration has resulted in user interface improvements and bug fixes that enhance the user experience for biologists.

GRNsight separates what users see (frontend), from how the data is handled (backend)



React migration resulted in tech stack improvements for developer experience and shorter build time

Feature	Classic	React / New Updates
Frontend	 Bootstrap  pug  jQuery 	 grommet  React 
Backend	 node  PostgreSQL express  JS	 node  PostgreSQL express  JS
Build Tool	 webpack	 Vite
Testing	 MOCHA  chai	 Vitest
State Management	Manual tracking of user interaction and input, and interface updates are tedious.	Automatic re-rendering of app based on user interaction and input, simplifying code.
Bug Fixes	Repetitive and complicated code makes bugs more likely and harder to find.	Modular code allows for easier discovery of bugs and fewer bugs created.

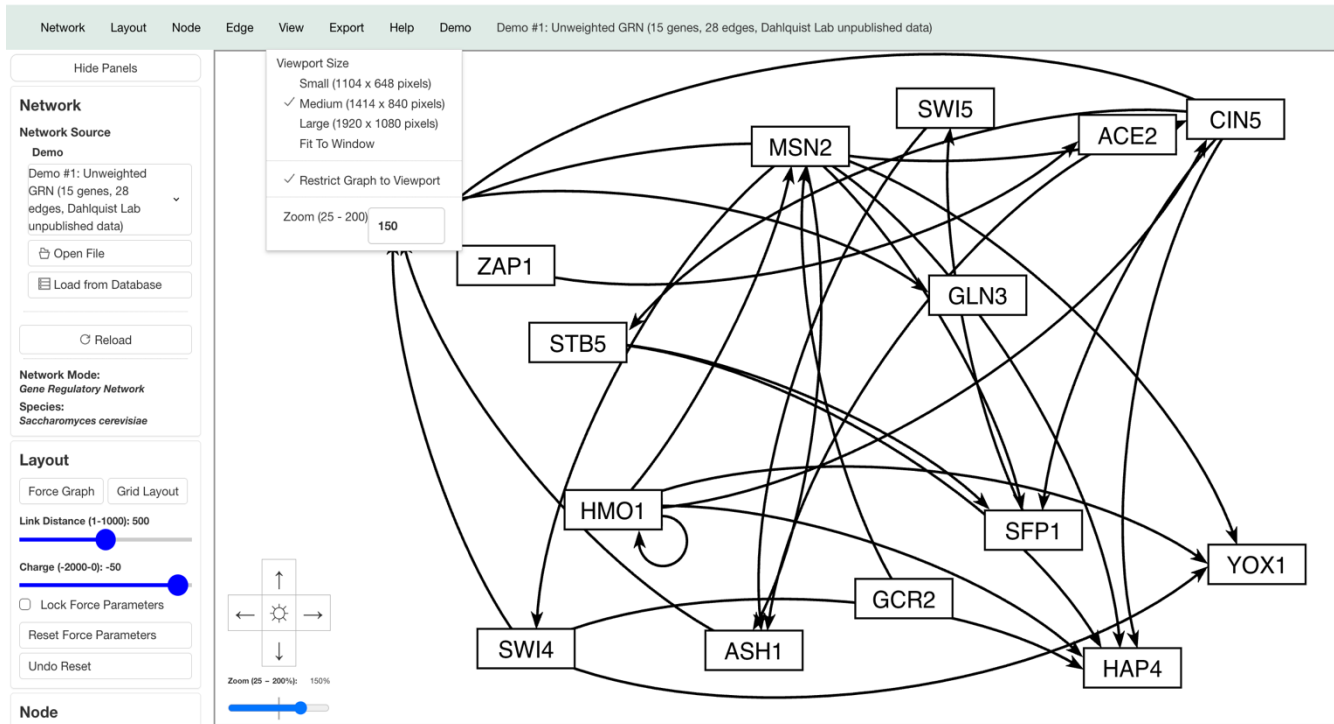
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GRNsight React went live on October 15, 2025, and can now display graphs

GRNsight React: v0.0.7 (04/09/2026)

Disclaimer: This version of GRNsight is currently under development and is unstable. For the most stable version, please go to the [GRNsight home page](#).



- <https://dondi.github.io/GRNsight/react-thesis-4081/>
- Viewport size, panning, and zoom are available graph features

React allows the fast implementation of collapsible side panels

GRNsight v7.4.1

Welcome to GRNsight. Please select a network to view.

NetworkLayoutNodeEdgeViewExportHelpDemo

Network

Network Source

Demo

Demo #1: Unweighted C ▾

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force GraphGrid Layout

Link Distance (1-1000): 500

Charge (-2000-0): -50

Lock Force Parameters

Reset Force Parameters

Undo Reset

Node

GRNsight React: v0.0.7 (04/09/2026)

Disclaimer: This version of GRNsight is currently under development and is unstable. For

NetworkLayoutNodeEdgeViewExportHelpDemo

Hide Panels

Network

Network Source

Demo

Demo #1: Unweighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data) ▾

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force GraphGrid Layout

Link Distance (1-1000): 500

Charge (-2000-0): -50

Lock Force Parameters

Reset Force Parameters

Undo Reset

Node

Enable Node Coloring

GRNsight React: v0.0.7 (04/09/2026)

Disclaimer: This version of GRNsight is currently under development and is unstable. For

NetworkLayoutNodeEdgeViewExportHelpDemo

Hide Panels

Network

Network Source

Demo

Demo #1: Unweighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data) ▾

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force GraphGrid Layout

Link Distance (1-1000): 500

Charge (-2000-0): -50

Lock Force Parameters

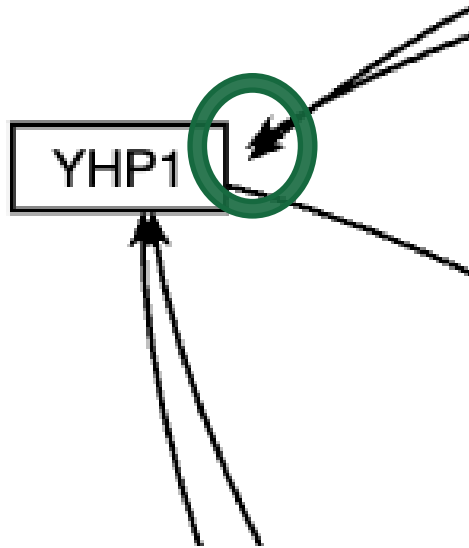
Reset Force Parameters

Undo Reset

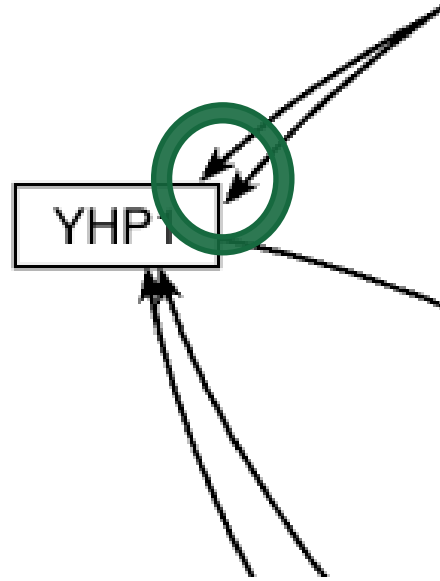
Node

Enable Node Coloring

The high-level code review facilitated the fixing of a long-standing arrowhead bug in GRNsight Classic



Classic Arrowheads with space between the perimeter



React Arrowheads touching the perimeter of the node

Summary

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Acknowledgements

Pictured Left to Right:

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Kam D. Dahlquist

Alex J. Miller

John David N. Dionisio

LMU Department of Biology

HHMI STEM Research Learning Community



<http://dondi.github.io/GRNsight/>



<https://github.com/dondi/GRNsight>

References

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- "PDB101: Molecule of the Month: Cyclin and Cyclin-dependent Kinase." *RCSB: PDB-101*, pdb101.rcsb.org/motm/236.
- Lee, T. I., Rinaldi, N. J., Robert, F., Odom, D. T., Bar-Joseph, Z., Gerber, G. K., ... & Young, R. A. (2002). Transcriptional regulatory networks in *Saccharomyces cerevisiae*. *science*, 298(5594), 799-804. <https://doi.org/10.1126/science.1075090>
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